

Feature

Ultra-low power start-up

- Cold start from 275 mV input voltage and 3 μ W input power (typical).

Very efficient energy extraction

- Open-circuit voltage sensing for Maximum Power Point Tracking (MPPT).
- Selectable open-circuit voltage ratios from 60% to 90% or fixed impedance.
- Programmable MPPT sensing period.
- MPPT voltage operation range from 100 mV to 4.5 V.

Adaptive and smart energy management

- DCDC switches automatically between boost, buck-boost and buck operation, according to input and output voltages, to maximize energy transfer.

Battery protection features

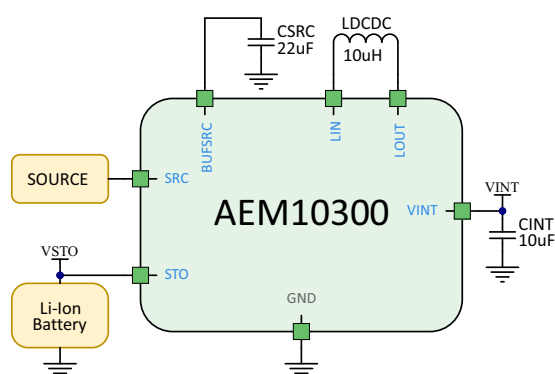
- Selectable over-charge and over-discharge protection for any type of rechargeable battery or (super-)capacitor.
- Fast super-capacitor charging.
- Dual cell super-capacitor balancing circuit.

Smallest footprint, smallest BOM

- Only three external components are required.
 - One 10 μ H inductor.
 - Two capacitors: 10 μ F and 22 μ F.

Applications

Smart home	Industrial sensor
Smart building	Retail
Edge IoT	PC accessories



Description

The AEM10300 is an integrated energy management circuit that extracts DC power from an ambient energy harvesting source to store energy in a storage element. The AEM10300 allows to extend battery lifetime and ultimately eliminates the primary energy storage element in a large range of applications.

Thanks to its Maximum Power Point Tracking system, the AEM10300 extracts the maximum energy available from the source. It integrates an ultra-low power DCDC converter which operates with input voltages ranging from 100 mV to 4.5 V.

With its unique cold start circuit, the AEM10300 can start harvesting with an input voltage as low as 275 mV and from an input power of 3 μ W. The preset protection levels determine the storage element voltages protection thresholds to avoid over-charging and over-discharging the storage element and thus, avoiding damaging it. Those are set through configuration pins. Moreover, custom threshold voltages can be obtained at the expense of a few configuration resistors.

The chip integrates all active elements for powering a typical wireless sensor. Only two capacitors and one inductor are required.

Device Information

Part Number	Package	Body size
10AEM10300C0000	QFN 28-pin	4x4mm

Evaluation Board

Part Number
2AAEM10300C001

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Figure 1: Simplified Schematic View

1. Introduction

The AEM10300 is a full-featured energy efficient power management circuit able to harvest energy from an energy source (connected to **SRC**) to charge a storage element (connected to **STO**). This is done with a minimal bill of material: only 2 capacitors and one inductor are needed for a basic setup.

The heart of the AEM10300 is a regulated switching DCDC converter with high power conversion efficiency.

At first start-up, as soon as a required cold-start voltage of 275 mV and a sparse amount of power of at least 3 μ W is available at the source, the AEM10300 coldstarts. After the cold start, the AEM extracts the power available from the source if the working input voltage is at least 100 mV.

Through four configuration pins (**STO_CFG[3:0]**), the user can select a specific operating mode out of 15 modes that cover most application requirements without any dedicated external component. Those operating modes define the protection levels of the storage element. If none of those 15 modes fit the user's storage element, a custom mode is also available to allow the user to define a mode with custom specifications.

Status pin **ST_STO** provides information about the voltage levels of the storage element.

ST_STO is HIGH when the voltage of the storage element V_{STO} is above V_{CHRDY} and is reset when the voltage drops below V_{OVDIS} .

The Maximum Power Point (MPP) ratio is configurable thanks to three configuration pins (**R_MPP[2:0]**) and ensures an optimum biasing of the harvester to maximize power extraction. Depending on the harvester, it is possible to adapt the timings of the MPP evaluations with the two configuration pins (**T_MPP[1:0]**) that sets the periodicity and the duration of the MPP evaluation.

The AEM10300's DCDC converter can work in two modes: **LOW POWER MODE** and **HIGH POWER MODE**, each one of these being optimized for a power range on **SRC**.

The charging of the storage element can be prevented by pulling **EN_STO_CH** to GND, typically to protect the storage element if the temperature is too low/high to safely charge it.

2. Pin Configuration and Functions

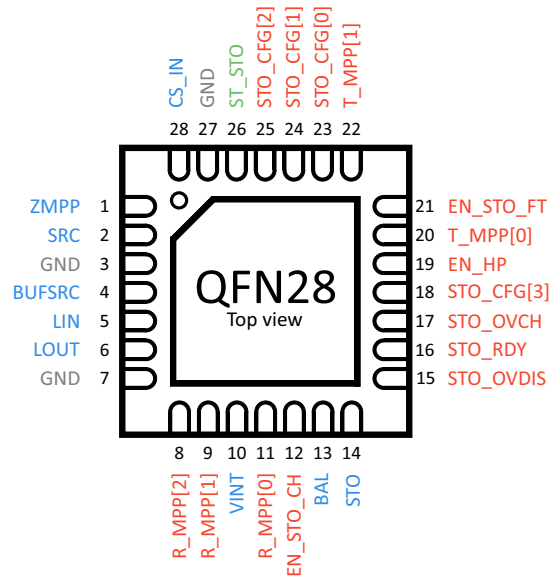


Figure 2: Pinout Diagram QFN 28-pin

NAME	PIN NUMBER	FUNCTION
Power Pins		
CS_IN	28	Input to connect an external cold start circuit.
ZMPP	1	Used for the configuration of the ZMPP (optional). Must be left floating if not used.
SRC	2	Connection to the harvested energy source.
BUFSRC	4	Connection to an external capacitor buffering the DCDC converter input.
LIN	5	DCDC inductance connection.
LOUT	6	DCDC inductance connection.
VINT	10	Internal voltage supply.
BAL	13	Connection to mid-point of a dual-cell supercapacitor (optional). Must be connected to GND if not used.
STO	14	Connection to the energy storage element - battery or (super-)capacitor. Cannot be left floating. Must be connected to a minimum capacitance of 100 μ F or to a rechargeable battery.

Table 1: Power and Status Pins



NAME	PIN NUMBER	LOGIC LEVEL		FUNCTION
		LOW	HIGH	
Configuration Pins				
R_MPP[0]	11	GND	VINT	Used to configure the MPPT ratio. Read as HIGH if left floating.
R_MPP[1]	9			
R_MPP[2]	8			
T_MPP[0]	20	GND	VINT	Used to configure the MPPT timings. Read as HIGH if left floating.
T_MPP[1]	22			
STO_CFG[0]	23	GND	VINT	Used to configure the storage element protection thresholds V_{OVDIS} , V_{CHRDY} and V_{OVCH} . Read as HIGH if left floating.
STO_CFG[1]	24			
STO_CFG[2]	25			
STO_CFG[3]	18			
STO_OVCH	17	Analog pins		Used to configure the storage element protection thresholds (V_{OVDIS} , V_{CHRDY} and V_{OVCH}) when STO_CFG[3:0] are set to custom mode (optional). Must be left floating if not used.
STO_RDY	16			
STO_OVDIS	15			
EN_STO_FT	21	GND	VINT	Used to configure the storage element feed-through feature. - HIGH: allows charges to flow directly from SRC to STO when SRC is above 5V. - LOW: normal operation. Read as HIGH if left floating.
EN_STO_CH	12	GND	VINT	Used to configure the charge of the storage element. - HIGH: storage element charge enabled. - LOW: storage element charge disabled. Read as HIGH if left floating.
EN_HP	19	GND	VINT	Used to configure the high power mode. - HIGH: HIGH POWER MODE enabled. - LOW: HIGH POWER MODE disabled. Read as HIGH if left floating.
Status Pins				
ST_STO	26	GND	LOAD	Logic output. - HIGH when V_{STO} rises above V_{CHRDY} . - LOW when V_{STO} drops below V_{OVDIS} .
Other				
GND	3, 7, 27			Ground connection, best possible connection to PCB ground plane.
	Exposed pad			

Table 2: Configuration and Ground Pins

3. Specifications

3.1. Absolute Maximum Ratings

Parameter		Min	Max	Unit
Operating junction temperature T_j		-40	85	°C
Storage temperature T_{stg}		-65	150	°C
Input voltage	STO, SRC, BUF SRC, LIN, LOUT, ZMPP, BAL, CS_IN	-0.3	5.5	V
	VINT, T_MPP[1:0], R_MPP[2:0], STO_CFG[3:0], STO_OVCH, STO_OVDIS, STO_RDY, EN_HP, EN_STO_CH	-0.3	2.75	V

Table 3: Absolute Maximum Ratings

3.2. ESD Ratings

Parameter		Value	Unit
Electrostatic discharge V_{ESD}	Human-Body Model (HBM) according to JEDEC JS001	2000	V
	Charged-Device Model (CDM) according to JEDEC JS002	500	V

Table 4: ESD ratings

ESD CAUTION	
	ESD (ELECTROSTATIC DISCHARGE) SENSITIVE DEVICE These devices have limited built-in ESD protection and damage may thus occur on devices subjected to high-energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality

3.3. Thermal Resistance

Package	θ_{JA}	θ_{JC}	Unit
QFN 28-pin 4x4mm	60	6	°C/W

Table 5: Thermal Resistance



3.4. Typical Electrical Characteristics at 25 °C

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Power Conversion						
P _{SRC,CS}	Minimum source power required for cold start	V _{STO} > V _{CHRDY}		3		μW
		V _{STO} < V _{CHRDY}		6		μW
V _{SRC,CS}	Minimum source input voltage required for cold start			275		mV
V _{SRC}	Input voltage of the energy source	During cold start	V _{SRC,CS}		4.5	V
		After cold start	0.1		4.5	V
R _{MPPT}	MPP Tracker module V _{MPP} / V _{OC} ratio.	See Table 10	60, 65, 70, 75, 80, 85 or 90, depending on R _{MPP} [2:0] configuration			%
Timing						
T _{MPPT,WAIT}	Wait time before V _{OC} measurement begins during MPP evaluation (see Section 5.4).		50% of Table 11		200% of Table 11	ms
T _{MPPT,MEASURE}	Duration of V _{OC} measurement during MPP evaluation.			0.82		ms
T _{MPPT,PERIOD}	Time between two MPP evaluations		50% of Table 11		200% of Table 11	s
T _{GPIO,MON}	GPIO monitoring rate			1.3		s
T _{STO,MON}	STO monitoring rate			167		μs
Storage element						
V _{OVCH}	Maximum voltage accepted on the storage element before disabling its charging	See Table 9	Depends on STO_CFG[3:0] configuration			V
V _{CHRDY}	Minimum voltage required on the storage element before asserting ST_STO					V
V _{OVDIS}	Minimum voltage accepted on the storage element before resetting ST_STO					V
V _{BAL,MIN}	Minimum voltage required on the BAL pin for the balancing circuit to operate			0.9		V
Internal supply & Quiescent Current						
V _{VINT}	Internal voltage supply			2.2		V
I _Q	Quiescent current on STO	V _{STO} = 3.7 V EN_HP = L		5.9		nA
Symbol	Logic Level		LOW		HIGH	
Logic output pins						
ST_STO	Logic output levels on the status STO pin		GND		V _{STO}	

Table 6: Typical Electrical Characteristics

3.5. Recommended Operating Conditions

Symbol	Parameter	Min	Typ	Max	Unit
External Components					
L_{DCDC}	Inductor of the DCDC converter		10		μH
C_{SRC}	Capacitor decoupling the SRC terminal	13 ¹	22		μF
C_{INT}	Capacitor decoupling the VINT terminal	5 ¹	10		μF
C_{STO}	Optional - Capacitor on STO if no storage element is connected (see Section 6.8.1)	100 ¹			μF
R_{ZMPP}	Optional - Used for the configuration of the ZMPP tracking function	10	Section 6.6	100K	Ω
STO_OVCH	Configuration of V_{OVCH} in custom mode	1	Section 6.2	100	$\text{M}\Omega$
STO_OVDIS	Configuration of V_{OVDIS} in custom mode				
STO_RDY	Configuration of V_{CHRDY} in custom mode				

Table 7: Recommended Operation Conditions

1. Consider all component tolerance and deratings. Typically, DC-bias derating has a major impact on capacitance on ceramic capacitors.



3.6. Typical Characteristics

3.6.1. DCDC Conversion Efficiency From SRC to STO in Low Power Mode

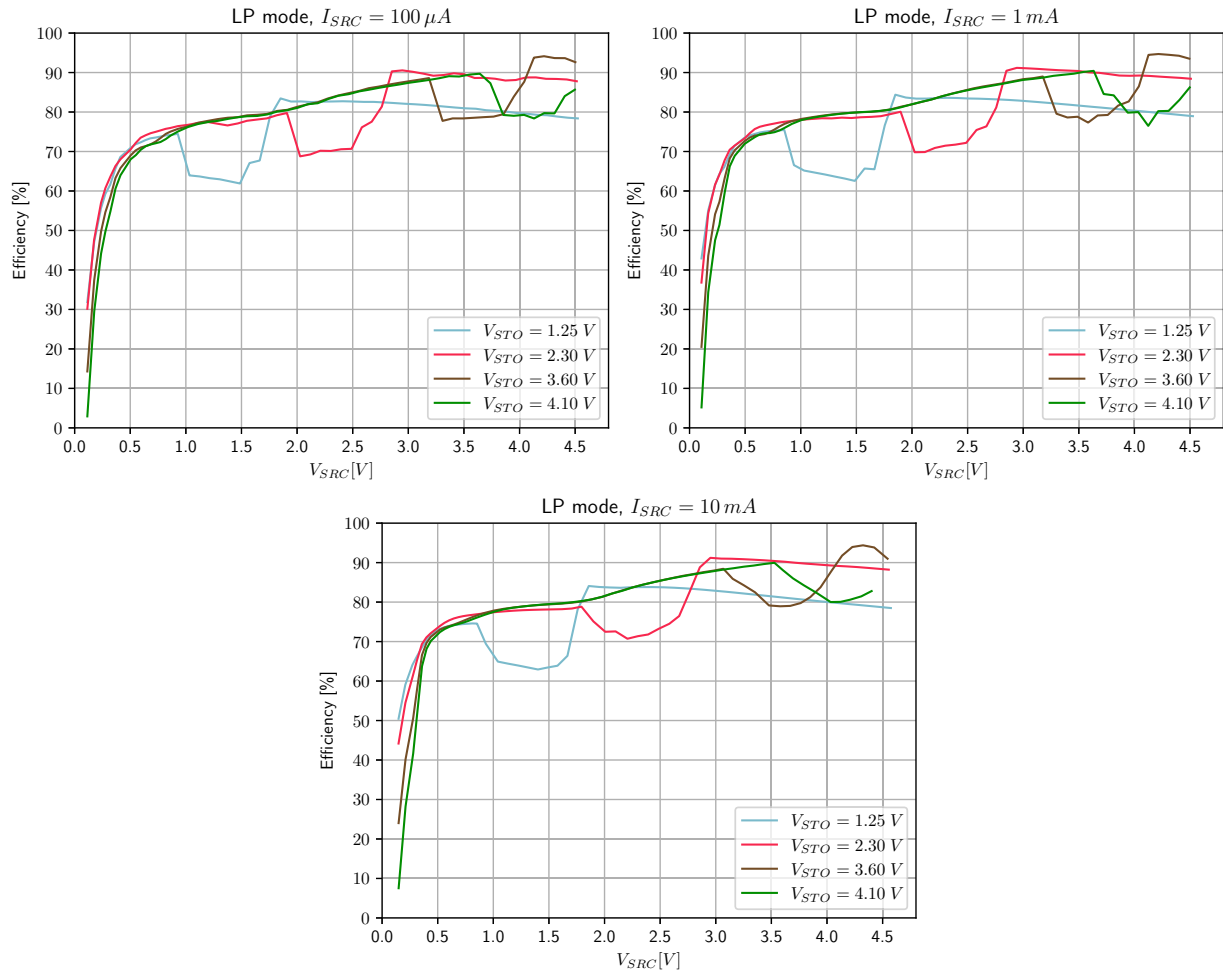


Figure 3: DCDC efficiency from SRC to STO in low power mode



3.6.2. DCDC Conversion Efficiency From SRC to STO in High Power Mode

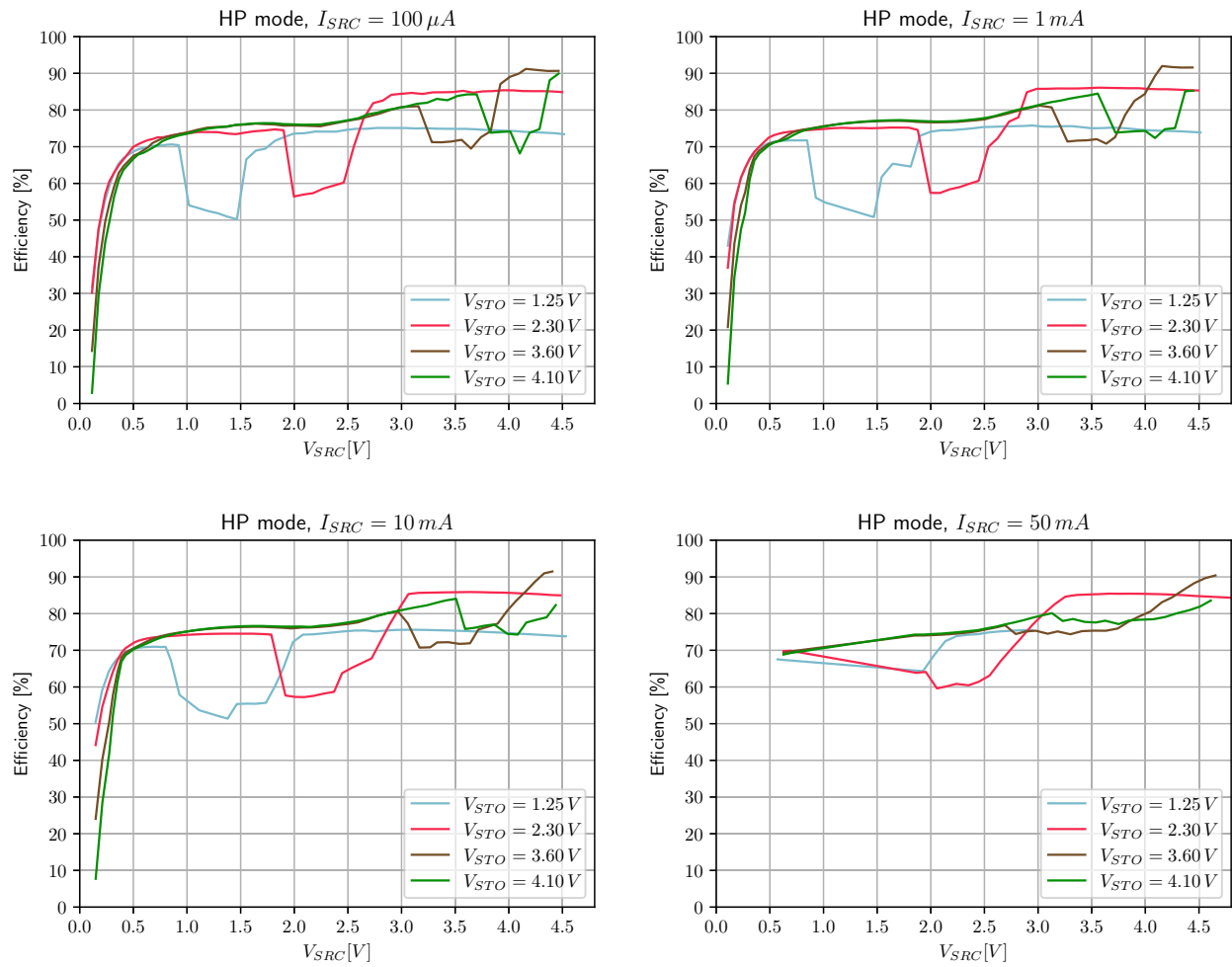


Figure 4: DCDC efficiency from SRC to STO in high power mode

4. Functional Block Diagram

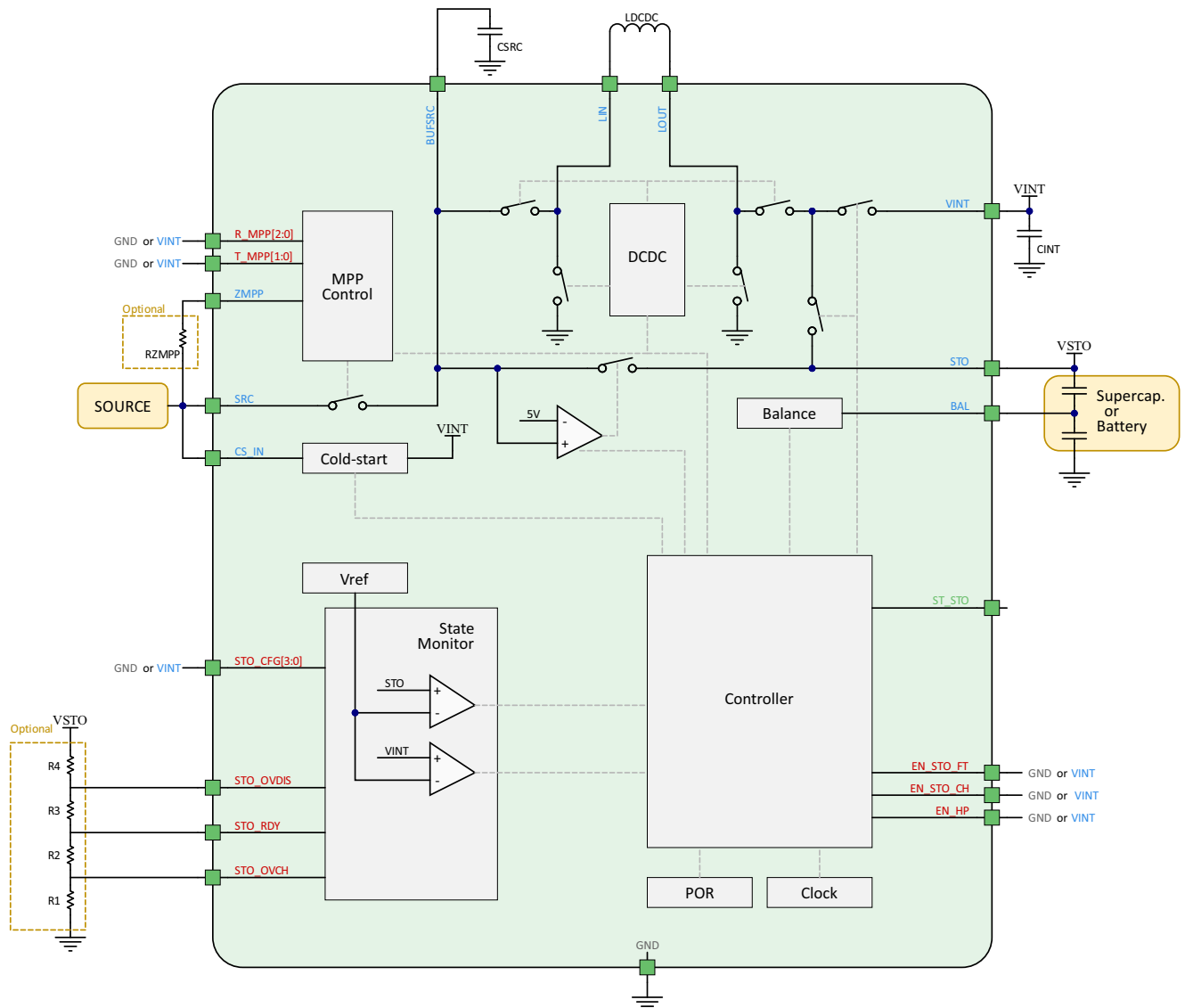


Figure 5: Functional Block Diagram

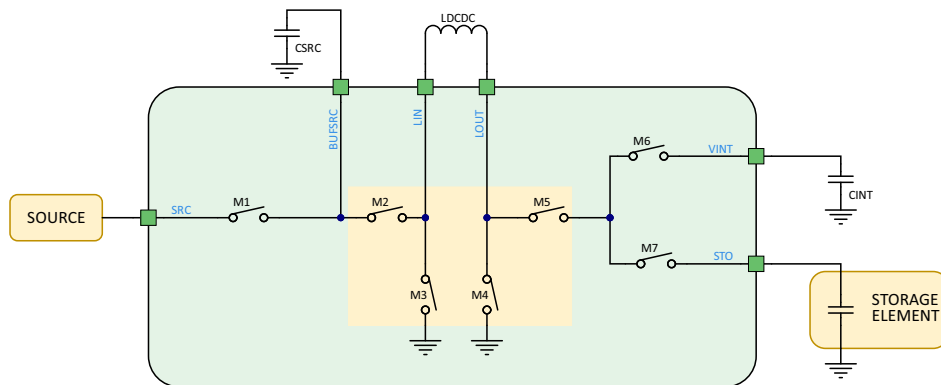


Figure 6: Simplified Schematic View of the AEM10300

5. Theory of Operation

5.1. DCDC Converter

The DCDC converter converts the voltage available at **BUFSRC** to a level suitable for charging the storage element **STO** or to regulate the internal supply **V_{VINT}**. The switching transistors of the DCDC converter are M2, M3, M4 and M5. Thanks to M6 and M7, the controller selects between **V_{VINT}** and **STO** respectively as the converter output. **STO** is selected as an output only when **V_{VINT}** does not need to be supplied.

The reactive power component of this converter is the external inductor **L_{DCDC}**. Periodically, the MPP control circuit disconnects the source from **BUFSRC** pin with the transistor M1 in order to let the harvester on **SRC** rise to its open-circuit voltage **V_{OC}** and measure it. This is done to define the optimal voltage level **V_{MPP}**, which is determined by applying the MPP ratio on **V_{OC}**. **BUFSRC** is decoupled by the capacitor **C_{SRC}**, which smooths the voltage against the current pulses pulled by the DCDC converter. The storage element is connected to the **STO** pin.

Depending on its input voltage and its output voltage, the DCDC converter will work as a boost converter, a buck converter or a buck-boost converter. The maximum power that can be harvested and supplied to the output depends on the power mode (**HIGH POWER MODE** or **LOW POWER MODE**), which is configured through the **EN_HP** pin (see Section 6.4).

DCDC Converter Mode	Input Voltage / Output Voltage
Boost	$V_{IN} < V_{OUT} - 250 \text{ mV}$
Buck	$V_{IN} > V_{OUT} + 250 \text{ mV}$
Buck - Boost	$V_{OUT} - 250 \text{ mV} < V_{IN} < V_{OUT} + 250 \text{ mV}$

Table 8: DCDC Converter Modes

5.2. Cold-Start Circuit

The AEM10300 is able to coldstart if the voltage on **CS_IN** is above 0.275 V. The minimum available power is:

- 3 μW if **V_{STO}** is above **V_{CHRDY}**.
- 6 μW if **V_{STO}** is below **V_{CHRDY}**.

CS_IN is typically connected to **SRC** to allow the AEM10300 to coldstart from the energy available on the harvester. Nevertheless, any other energy source can be connected to **CS_IN** as long as it meets the electrical specifications constraints described in Sections 3.4 and 3.5.

5.3. AEM10300 States Description

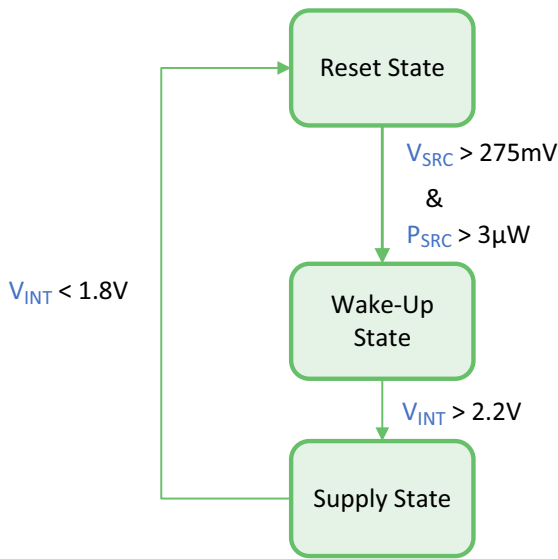


Figure 7: Diagram of the AEM10300 States

5.3.1. Reset and Wake Up States

The **RESET STATE** is a state where all nodes are deeply discharged and there is no available energy to be harvested. As soon as the required cold start voltage of 275 mV and a sparse amount of power of just 3 μ W become available on **CS_IN** (usually connected to **SRC**), the AEM10300 switches to **WAKE-UP STATE**, and energy is extracted from **SRC** to make V_{INT} rise to 2.2 V. When V_{INT} reaches 2.2 V, the AEM10300 switches to **SUPPLY STATE**.

5.3.2. Supply State

In **SUPPLY STATE**, three scenarios are possible:

- There is enough power provided by the source (**SRC**) to keep V_{INT} at 2.2 V and to charge the storage element. When the voltage on **VINT** falls below its 2.2 V regulation set point because of the AEM10300 internal circuit current consumption discharging C_{INT} , the DCDC converter switches its output to recharge C_{INT} from **SRC**, thus, keeping V_{INT} regulated. The rest of the time, the power on source is used to charge the storage element on **STO**. In this case, the circuit remains in **SUPPLY STATE**. If all nodes are fully charged, the DCDC converter is disabled to prevent over-charging the storage element, and the **SRC** pin is set to high impedance.

- Due to a lack of power from the source, V_{STO} falls below V_{OVDIS} . In this case, **ST_STO** is set LOW and the AEM10300 remains in **SUPPLY STATE**. **ST_STO** will stay LOW until V_{STO} reaches V_{CHRDY} again.
- There is not enough power on **SRC** to maintain V_{INT} to 2.2 V. In this case, the circuit enters in **RESET STATE**.

5.4. Maximum Power Point Tracking

In **SUPPLY STATE**, the voltage on **SRC** is regulated by an internal Maximum Power Point Tracking (MPPT) module. The MPPT module evaluates V_{MPP} , the voltage at which the source provides the highest possible power, as a given fraction of the open-circuit voltage of the source (V_{OC}). This ratio is set by the **R_MPP[2:0]** terminals according to Table 10. The AEM10300 supports any V_{MPP} levels in the range from 100 mV to 4.5 V. It offers a choice of seven values for the V_{MPP} / V_{OC} fraction. It can also match the input impedance of the DCDC converter with an impedance connected to the **ZMPP** terminal as explained in Section 6.6.

The MPPT module evaluates periodically the open-circuit voltage (V_{OC}) with the following sequence to ensure optimal power extraction at any time:

- The AEM10300 stops extracting power from the **SRC** during $T_{MPPT, WAIT}$ to allow the **SRC** voltage to rise to V_{OC} .
- Once this delay elapses, the AEM10300 performs the measurement of V_{OC} during $T_{MPPT, MEASURE}$.
- After the measurement, the AEM10300 resumes extracting power by regulating the **SRC** voltage to the newly determined V_{MPP} .
- This MPP evaluation is repeated every $T_{MPPT, PERIOD}$.

$T_{MPPT, WAIT}$ and $T_{MPPT, PERIOD}$ are set according to Table 11 by the **T_MPP[1:0]** terminals while $T_{MPPT, MEASURE}$ is constant for any configuration (see Table 6).

The total time during which the AEM10300 does not extract power correspond to the sum of $T_{MPPT, WAIT}$ and $T_{MPPT, MEASURE}$ (see Figure 8).

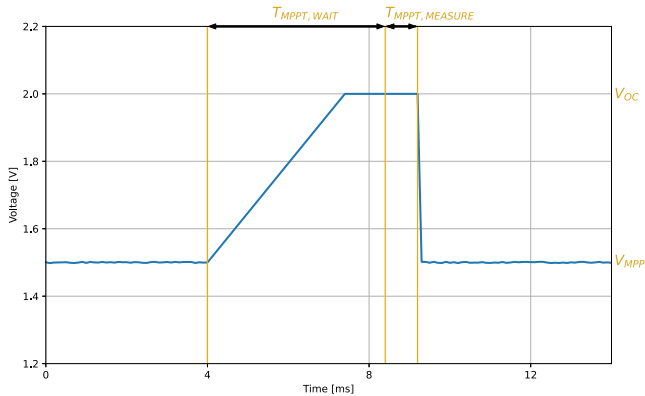


Figure 8: MPPT evaluation behavior (with $T_{MPPT, WAIT}$ of 4.4 ms)

5.5. Balancing for Dual-Cell Supercapacitor

The AEM10300 balancing feature allows to balance the internal voltage of a dual-cell supercapacitor connected to **STO** in order to avoid damaging the supercapacitor because of excessive voltage on one cell.

If a battery, a capacitor or a single-cell supercapacitor is connected on **STO**, the balancing feature is not needed. In this case, **BAL** must be connected to **GND** to disable the balancing.

If a dual-cell supercapacitor is connected on **STO**, **BAL** must be connected to the node between both cells of the supercapacitor. The balancing circuit will compensate for any mismatch of the two cells that could lead to the overcharge of one of two cells. The balancing circuit ensures that **BAL** remains close to $V_{STO} / 2$.

The balancing circuit works as follows:

- The balancing circuit is enabled only when $V_{BAL} > V_{BAL, MIN}$ (see Table 6).
- If $V_{BAL} > \frac{V_{STO}}{2}$: the AEM10300 enables a switch between **BAL** and **GND** to discharge the bottom supercapacitor cell to **GND** (up to 20 mA).
- If $V_{BAL} < \frac{V_{STO}}{2}$: the AEM10300 enables a switch between **STO** and **BAL** to discharge the top supercapacitor cell to the bottom supercapacitor cell (up to 20 mA).

6. System Configuration

6.1. Storage Element Configuration

Through four configuration pins (**STO_CFG[3:0]**), the user can set a particular operating mode from a range that covers most application requirements, without any dedicated external component as shown in Table 9. The three threshold levels are defined as:

- **V_{OVCH}**: maximum voltage accepted on the storage element before disabling its charging.
- **V_{CHRDY}**: minimum voltage required on the storage element before setting **ST_STO** HIGH.
- **V_{OVDIS}**: minimum voltage accepted on the storage element before considering the storage element as depleted, thus, setting **ST_STO** LOW.

DISCLAIMER: storage element thresholds provided in the table below are indicative to support a wide range of storage element variants. They are provided as is to the best knowledge of e-peas's application laboratory. They should not replace the actual values provided in the storage element manufacturer's specifications and datasheet.

Configuration pins				Storage element threshold voltages			Typical use
STO_CFG[3:0]				V_{OVDIS}	V_{CHRDY}	V_{OVCH}	
L	L	L	L	3.00 V	3.50 V	4.05 V	LiCoO ₂ battery, Li-Po battery, Lithium Titanate (3.8 V) battery (long life).
L	L	L	H	2.80 V	3.10 V	3.60 V	LiFePO ₄ battery, Lithium capacitor (LiC).
L	L	H	L	1.85 V	2.40 V	2.70 V	Dual-cell NiMH battery, Lithium-Titanate (2.4V) battery.
L	L	H	H	0.20 V	1.00 V	4.65 V	Dual-cell supercapacitor.
L	H	L	L	0.20 V	1.00 V	2.60 V	Single-cell supercapacitor.
L	H	L	H	1.00 V	1.20 V	2.95 V	Single-cell supercapacitor.
L	H	H	L	1.85 V	2.30 V	2.60 V	Lithium-Titanate battery (2.4V).
L	H	H	H	Custom Mode (single-cell NiMH battery, LiC, etc.) ¹ .			
H	L	L	L	1.10 V	1.25 V	1.50 V	Ni-Cd single-cell battery.
H	L	L	H	2.20 V	2.50 V	3.00 V	Ni-Cd dual-cell battery.
H	L	H	L	1.45 V	2.00 V	4.65 V	Dual-cell supercapacitor.
H	L	H	H	1.00 V	1.20 V	2.60 V	Single-cell supercapacitor.
H	H	L	L	2.00 V	2.30 V	2.60 V	Solid State battery.
H	H	L	H	3.00 V	3.50 V	4.35 V	LiCoO ₂ battery, Li-Po battery, Lithium Titanate (3.8 V) battery.
H	H	H	L	2.60 V	2.70 V	4.00 V	Tadiran TLI.
H	H	H	H	2.60 V	3.50 V	3.90 V	Tadiran HLC.

Table 9: Storage Element Configuration Pins

1. An example of a single-cell NiMH batteries optimized custom mode setting can be found at Section 7.2.

6.2. Custom Mode Configuration

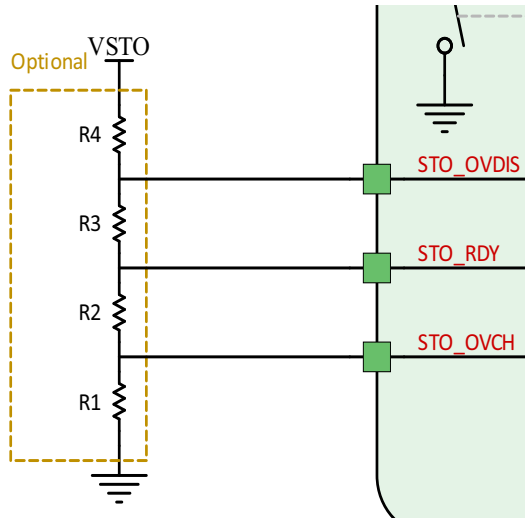


Figure 9: Custom Mode Settings

When **STO_CFG[3:0] = LHHH**, the custom mode is selected and all four configuration resistors must be wired as shown in Figure 9. V_{OVCH} , V_{CHRDY} and V_{OVDIS} are defined thanks to R1, R2, R3 and R4, which can be determined within the following constraints:

- $R_T = R_1 + R_2 + R_3 + R_4$
- $1\text{M}\Omega \leq R_T \leq 100\text{M}\Omega$
- $R_1 = R_T \cdot \frac{1\text{V}}{V_{OVCH}}$
- $R_2 = R_T \cdot \left(\frac{1\text{V}}{V_{CHRDY}} - \frac{1\text{V}}{V_{OVCH}} \right)$
- $R_3 = R_T \cdot \left(\frac{1\text{V}}{V_{OVDIS}} - \frac{1\text{V}}{V_{CHRDY}} \right)$
- $R_4 = R_T \cdot \left(1 - \frac{1\text{V}}{V_{OVDIS}} \right)$

The resistors should have high values to make the additional power consumption negligible. Moreover, the following constraints must be respected to ensure the functionality of the chip:

- $V_{CHRDY} + 0.05\text{V} \leq V_{OVCH} \leq 4.5\text{V}$
- $V_{OVDIS} + 0.05\text{V} \leq V_{CHRDY} \leq V_{OVCH} - 0.05\text{V}$
- $1\text{V} \leq V_{OVDIS}$

6.3. Disable Storage Element Charging

Pulling down **EN_STO_CH** pin to **GND** disables the charging of the storage element connected to **STO**. This can be done for example to protect the storage element when the system detects that the environment temperature is too low or too high to safely charge the storage element.

While **EN_STO_CH** is pulled down, **VINT** can still be supplied from **SRC**.

To enable charging the storage element on **STO**, **EN_STO_CH** must be pulled up to **VINT** or left floating (pin is pulled up internally).

6.4. High Power / Low Power Mode

When **EN_HP** is pulled to **VINT**, the DCDC converter is configured to **HIGH POWER MODE**. This allows higher currents to be extracted from the DCDC converter input (**SRC**) to the DCDC converter output (**STO**) with a trade-off in efficiency.

6.5. MPPT Configuration

The MPPT module is configured through the following pins:

- **R_MPP[2:0]** allows for configuring the V_{MPP} / V_{OC} tracking ratio, that has to be chosen according to the characteristics of the source. Please note that if the selected mode is **ZMPP**, an external resistor must be added to the circuit as explained in Section 6.6.
- **T_MPP[1:0]** allows for configuring the duration of an MPP evaluation and the time between two MPP evaluations.

Configuration pins			MPPT ratio
R_MPP[2:0]			V_{MPP} / V_{OC}
L	L	L	60%
L	L	H	65%
L	H	L	70%
L	H	H	75%
H	L	L	80%
H	L	H	85%
H	H	L	90%
H	H	H	ZMPP

Table 10: MPPT Ratio Configuration Pins

Configuration pins		MPPT timing	
T_MPP[1:0]		$T_{MPPT, WAIT} + T_{MPPT, MEASURE}^1$	$T_{MPPT, PERIOD}$
L	L	5.19 ms	280 ms
L	H	70.8 ms	4.5 s
H	L	280 ms	17.9 s
H	H	1.12 s	71.7 s

Table 11: MPPT Timing Configuration Pins

1. The total duration of the open-circuit voltage evaluation is the sum of $T_{MPPT, WAIT}$ and $T_{MPPT, MEASURE}$. $T_{MPPT, MEASURE}$ is fixed, see Table 6).

6.6. ZMPP Configuration

Instead of working at a ratio of the open-circuit voltage, the AEM10300 can regulate the input impedance of the DCDC converter so that it matches a constant impedance R_{ZMPP} connected to the **ZMPP** pin (see Figure 10). In this case, the AEM10300 regulates V_{SRC} at a voltage that is the product of the **ZMPP** resistance R_{ZMPP} and the current available at the **SRC** input.

$$10 \Omega \leq R_{ZMPP} \leq 100 \text{ k}\Omega$$

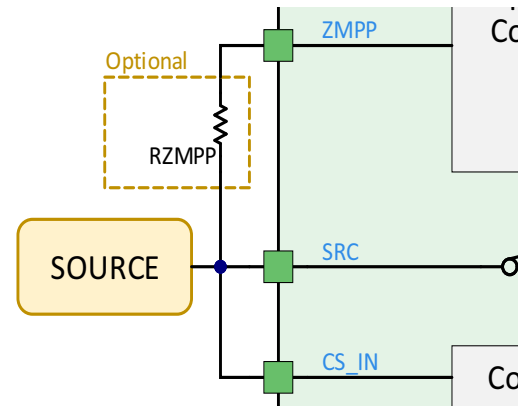


Figure 10: R_{ZMPP} Connection to the AEM10300

6.7. Source to Storage Element Feed-Through

When the harvester connected to **SRC** delivers a high amount of power, the AEM might not be able to pull enough current to regulate V_{SRC} to the MPP voltage, thus, increasing the voltage on **SRC**, eventually above 5V. To maximize the energy extracted in that case, the AEM10300 can be configured to create a direct feed-through current path from **SRC** to **STO** when V_{SRC} is above 5V. This is measured when the AEM is pulling current from the source (not during an MPP evaluation).

If the MPPT module detects that V_{SRC} is higher than 4 V and **EN_STO_FT** is set HIGH, the **SRC** is monitored. From that moment, if the AEM10300 detects that V_{SRC} rises above 5 V and if the storage element is not fully charged, the switch between **SRC** and **STO** pins is closed until V_{SRC} drops below 5 V or until the storage element is fully charged.

This feature is enabled by pulling up **EN_STO_FT** pin to **VINT**. However, it is disabled if the storage element is fully charged, or when a MPP evaluation is occurring. Therefore the circuit must still be protected from any overshoot voltage on **SRC** above 5.5 V, for instance by a zener diode.

6.8. External Components

Refer to Figure 16 to have an illustration of the external components wiring.

6.8.1. Storage Element Information

The energy storage element of the AEM10300 can be a rechargeable battery, a supercapacitor or a capacitor. The size of the storage element must be determined so that its voltage does not fall below V_{OVDIS} even during current peaks pulled by the application. If the internal resistance of the storage element cannot sustain this voltage limit, it is advisable to decouple the battery with a capacitor.

If the application expects a disconnection of the battery (e.g. because of a user removable connector), the PCB must include a capacitor C_{STO} of at least 100 μF connected between **STO** and **GND**. The leakage current of the storage element should be small as leakage currents directly impact the quiescent current of the whole subsystem.

6.8.2. External Inductor Information

The AEM10300 operates with one standard miniature inductor. L_{DCDC} must sustain a peak current of at least 1 A and a switching frequency of at least 10 MHz. Low equivalent series resistance (ESR) favors the power conversion efficiency of the DCDC converter. The recommended value is 10 μH .

6.8.3. External Capacitors Information

6.8.3.1. C_{SRC}

This capacitor acts as an energy buffer at the input of the DCDC converter. It prevents large voltage fluctuations when the DCDC converter is switching. The recommended nominal value is 22 μF .

6.8.3.2. C_{INT}

This capacitor acts as an energy buffer for the internal voltage supply. The recommended nominal value is 10 μF .

7. Typical Application Circuits

7.1. Example Circuit 1

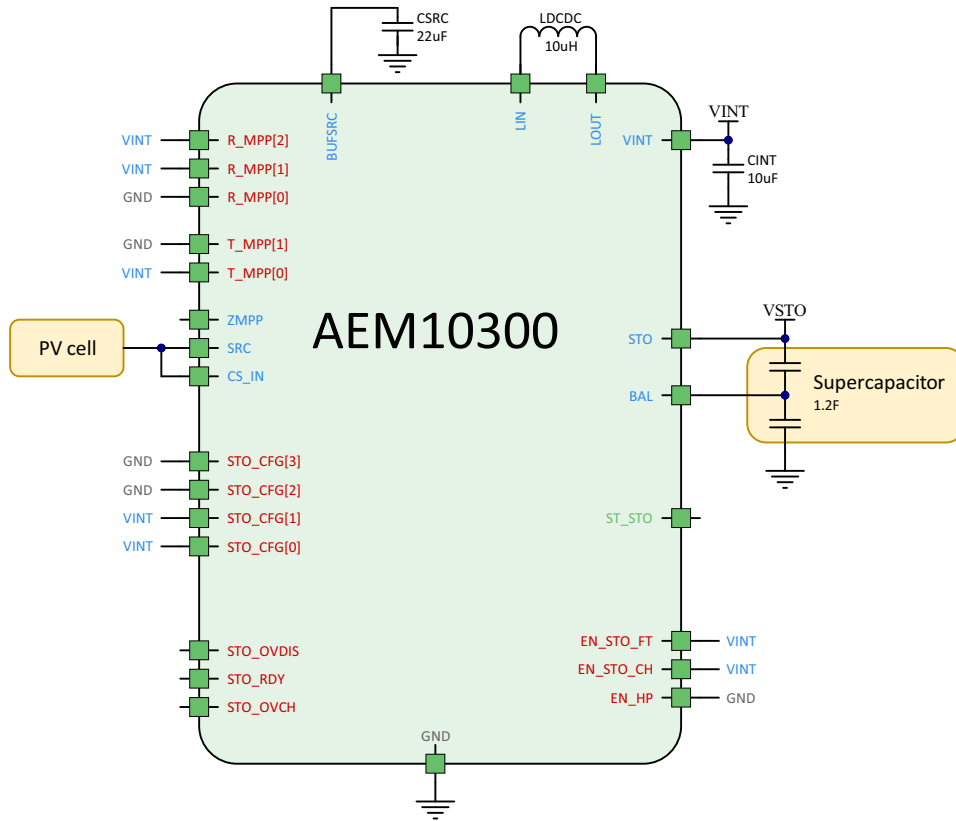


Figure 11: Typical Application Circuit 1

The circuit is an example of a system with solar energy harvesting. It uses a pre-defined operating mode that uses standard components, and a supercapacitor as energy storage.

- Energy source: PV cell.
- **R_MPP[2:0]** = HHL: the MPP tracker ratio is set to 90%.
- **T_MPP[1:0]** = LH: the MPP evaluation period is 4.5 s and the MPP wait duration is 70.8 ms.
- **STO_CFG[3:0]** = LLHH: the storage element is a dual-cell supercapacitor, with:
 - $V_{OVCH} = 4.65\text{ V}$
 - $V_{CHRDY} = 1.00\text{ V}$
 - $V_{OVDIS} = 0.20\text{ V}$
- The balancing pin of the dual-cell supercapacitor is connected to **BAL**.
- **EN_STO_CH** is connected to **VINT**: the charging of the storage element on **STO** is enabled.
- **EN_HP** is connected to **GND**: the DCDC converter is in **LOW POWER MODE**.
- **EN_STO_FT** is connected to **VINT**: **SRC** to **STO** feed-through is enabled when V_{SRC} is above 5 V.

7.2. Example Circuit 2

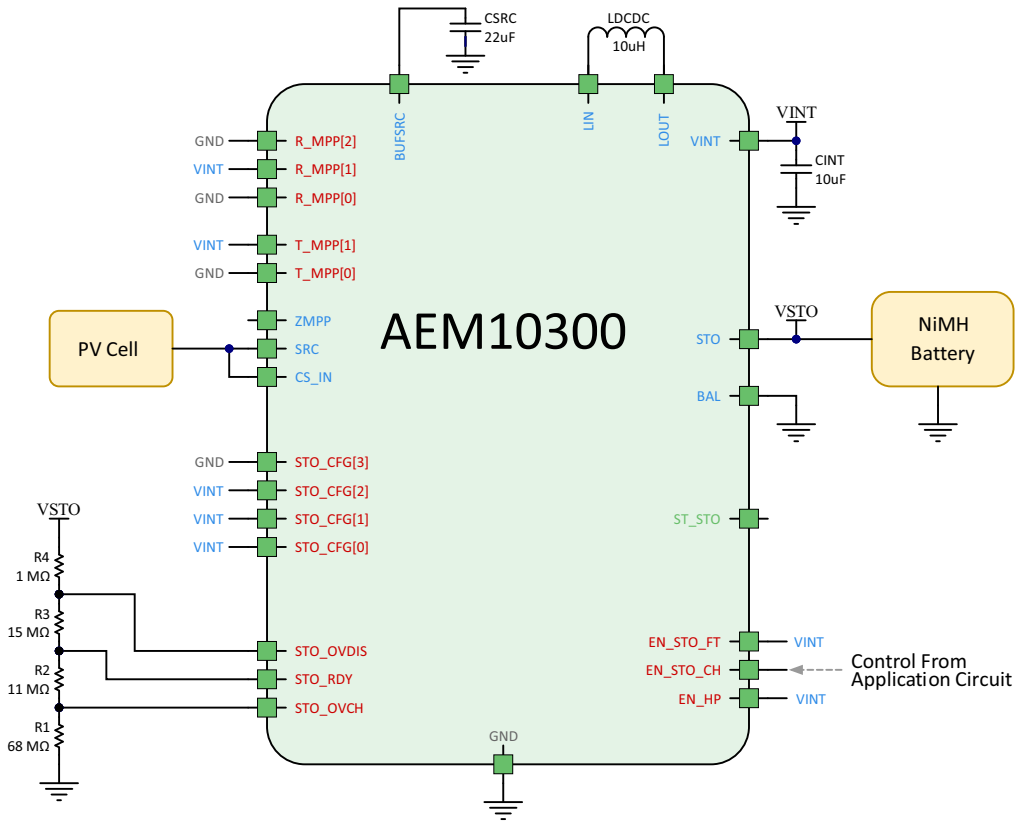


Figure 12: Typical Application Circuit 2

The circuit is an example of a system with solar energy harvesting. It uses a rechargeable NiMH battery as storage element. The voltage thresholds are set by the custom mode.

- Energy source: PV cell.
- **R_MPP[2:0]** = LHL: the MPP ratio is set to 70%.
- **T_MPP[1:0]** = HL: the MPP evaluation period is 17.9 s and the MPP wait duration is 280 ms.
- **STO_CFG[3:0]** = LHHH: the storage element is a NiMH rechargeable battery, used with custom mode:
 - $V_{OVDIS} = 1.00 \text{ V}$
 - $V_{CHRDY} = 1.20 \text{ V}$
 - $V_{OVCH} = 1.40 \text{ V}$
- Custom mode resistor divider calculations (values have been chosen to match E24 series value):
 - $R_T = 95 \text{ M}\Omega$
 - $R_1 = R_T \cdot \frac{1\text{V}}{V_{OVCH}} \approx 68 \text{ M}\Omega$
 - $R_2 = R_T \cdot \left(\frac{1\text{V}}{V_{CHRDY}} - \frac{1\text{V}}{V_{OVCH}} \right) \approx 11 \text{ M}\Omega$

$$- R_3 = R_T \cdot \left(\frac{1\text{V}}{V_{OVDIS}} - \frac{1\text{V}}{V_{CHRDY}} \right) \approx 15 \text{ M}\Omega$$

$$- R_4 = R_T \cdot \left(1 - \frac{1\text{V}}{V_{OVDIS}} \right) \approx 1 \text{ M}\Omega$$

- **BAL** is not used (not a dual-cell storage element) so it is connected to **GND**.
- **EN_STO_CH**: the charging of the storage element present on **STO** is controlled by an external application circuit, typically by a micro-controller GPIO output.
- **EN_HP** is connected to **VINT**: the DCDC converter is in **HIGH POWER MODE**.
- **EN_STO_FT** is connected to **VINT**: **SRC** to **STO** feed-through is enabled when V_{SRC} is above 5 V.

*NOTE: for LiC (Lithium-ion Capacitor) storage elements, or others that would not be covered by **STO_CFG[3:0]** presets, please apply the same equations as in the above example to determine custom mode resistors values. E24 series values for typical storage elements can be found in the AEM10300 Configuration Tool spreadsheet, to be downloaded on e-peas website.*

8. Circuit Behavior

8.1. Wake-up state and Supply state

8.1.1. Configuration

- $STO_CFG[3:0] = LLLL$
 - $V_{OVDIS} = 3.00\text{ V}$
 - $V_{CHRDY} = 3.50\text{ V}$
 - $V_{OVCH} = 4.05\text{ V}$
- $R_MPP[2:0] = LLL$ (60%)
- $T_MPP[1:0] = LL$ (5.19 ms / 280 ms)
- $C_{STO} = 10\text{ mF}$
- SRC : 5 mA current source with 3 V voltage compliance
- $EN_HP = H$ (high power mode)
- $EN_STO_CH = H$ (storage element charge enabled)

8.1.2. Observations

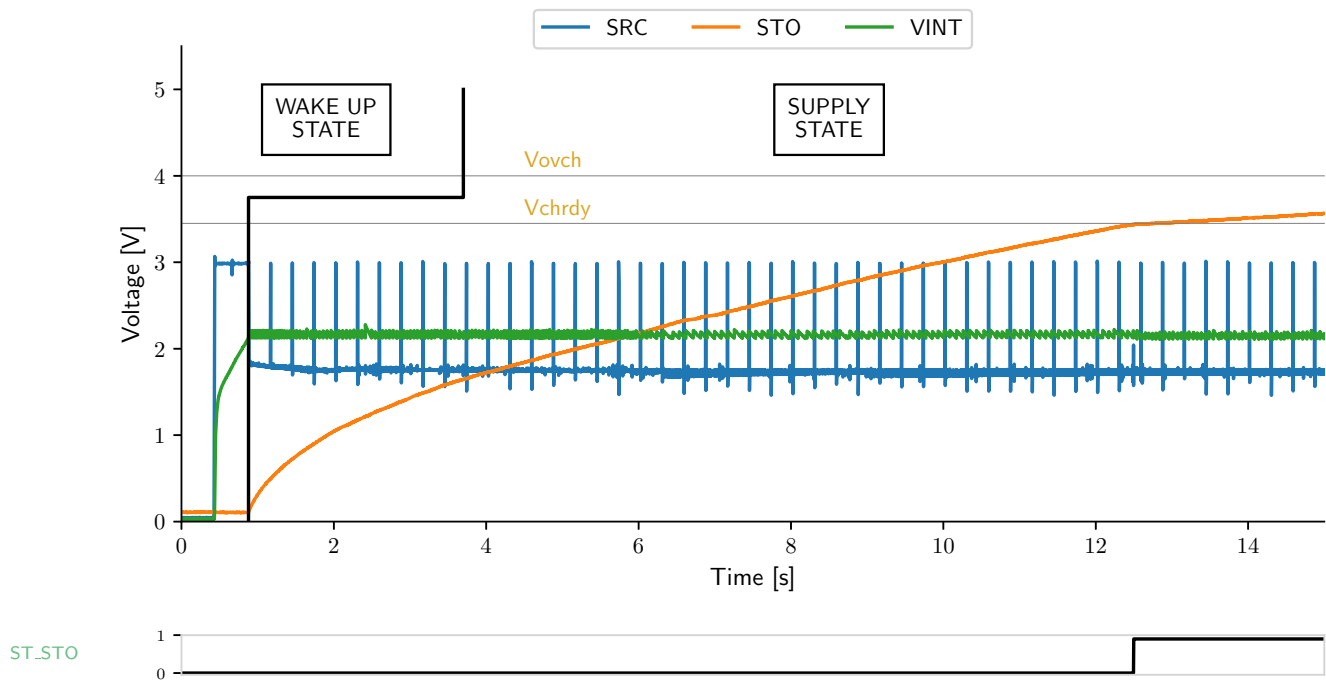


Figure 13: Wake-up state and Supply state

- The AEM10300 is initially in **WAKE-UP STATE**. ST_STO is LOW, the storage element connected on STO is depleted and no energy is applied on SRC .
- Once the supply connected on SRC is switched on, the AEM10300 coldstarts. V_{VINT} rises around 2.2 V, the AEM10300 switches to **SUPPLY STATE**.
- In **SUPPLY STATE**, the AEM10300 charges the storage element connected on STO . V_{SRC} is regulated to V_{MPP} thanks to the MPPT module. The open circuit voltage V_{OC} is measured every $T_{MPPT,PERIOD}$.
- Once V_{STO} reaches V_{CHRDY} , ST_STO is asserted.

8.2. Supply state

8.2.1. Configuration

- $STO_CFG[3:0] = LLLL$
 - $V_{OVDIS} = 3.00\text{ V}$
 - $V_{CHRDY} = 3.50\text{ V}$
 - $V_{OVCH} = 4.05\text{ V}$
- $R_MPP[2:0] = LLL$ (60%)
- $T_MPP[1:0] = LL$ (5.19 ms / 280 ms)
- $C_{STO} = 10\text{ mF}$
- SRC : 5 mA current source with 3 V voltage compliance
- $EN_HP = H$ (high power mode)
- $EN_STO_CH = H$ (storage element charge enabled)
- 1 k Ω between STO and GND , connected between 7 s and 25.5 s (no load on STO the rest of the time)

8.2.2. Observations

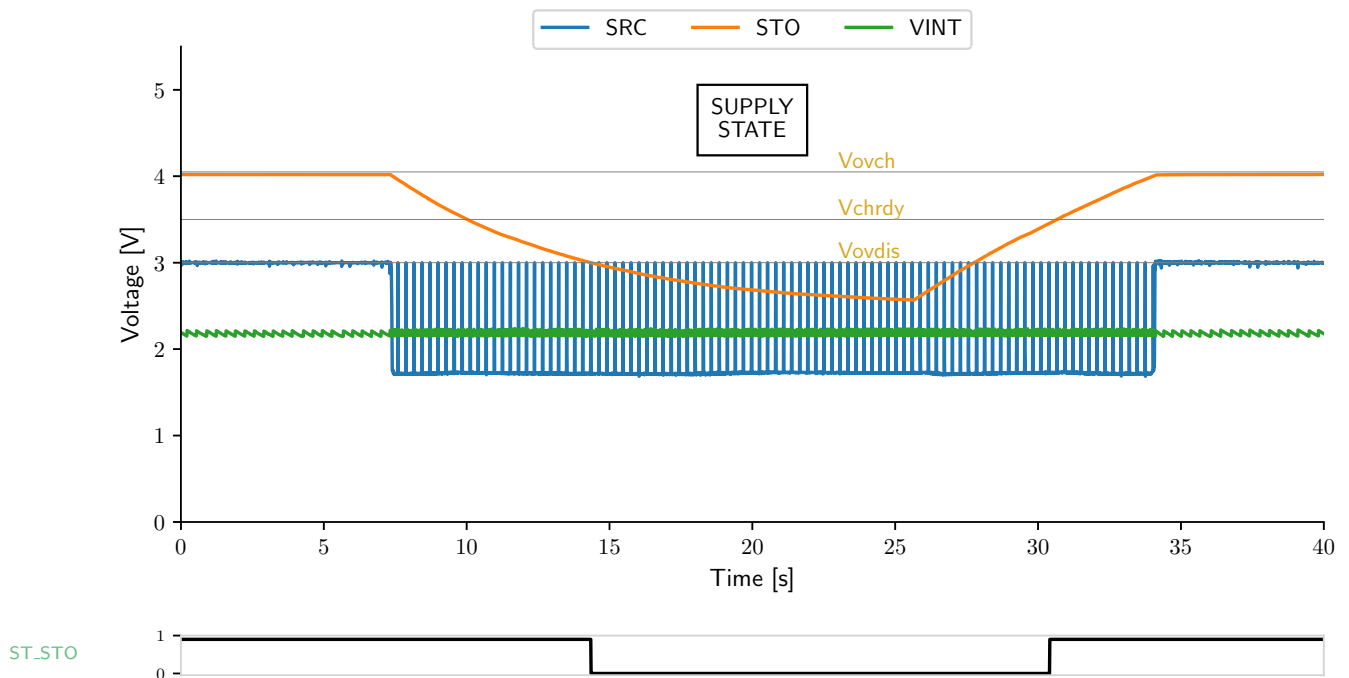


Figure 14: Supply state

- The AEM10300 is in **SUPPLY STATE** and V_{VINT} is regulated around 2.2 V the whole time.
- Before 7 s, V_{STO} is above V_{OVCH} , no energy is harvested and the SRC pin is set to high impedance. V_{SRC} is equal to the open-circuit voltage V_{OC} .
- After 7 s, a load is connected on STO , causing V_{STO} to discharge below V_{OVCH} . The AEM10300 starts harvesting energy on SRC by regulating V_{SRC} to V_{MPP} thanks to the MPPT module that measure V_{OC} every $T_{MPPT,PERIOD}$. Since the power drained by the load is higher than the harvested power, V_{STO} keeps discharging.
- After 25.5 s, the load on STO is removed. The harvested power on SRC allows recharging V_{STO} .
- ST_STO is set LOW once V_{STO} drops below V_{OVDIS} . It is set HIGH again once V_{STO} reaches V_{CHRDY} .
- Once V_{STO} reaches V_{OVCH} , the AEM10300 stops charging the storage element connected on STO and the SRC pin is set to high impedance.

8.3. Supply state and Reset state

8.3.1. Configuration

- **STO_CFG[3:0]** = LLLL
 - **V_{OVDIS}** = 3.00 V
 - **V_{CHRDY}** = 3.50 V
 - **V_{OVCH}** = 4.05 V
- **R_MPP[2:0]** = LLL (60%)
 - **T_MPP[1:0]** = LL (5.19 ms / 280 ms)
 - **C_{STO}** = 10 mF
 - **SRC**: 5 mA current source with 3 V voltage compliance (stopped after 19 seconds)
 - **EN_HP** = H (high power mode)
 - **EN_STO_CH** = H (storage element charge enabled)

8.3.2. Observations

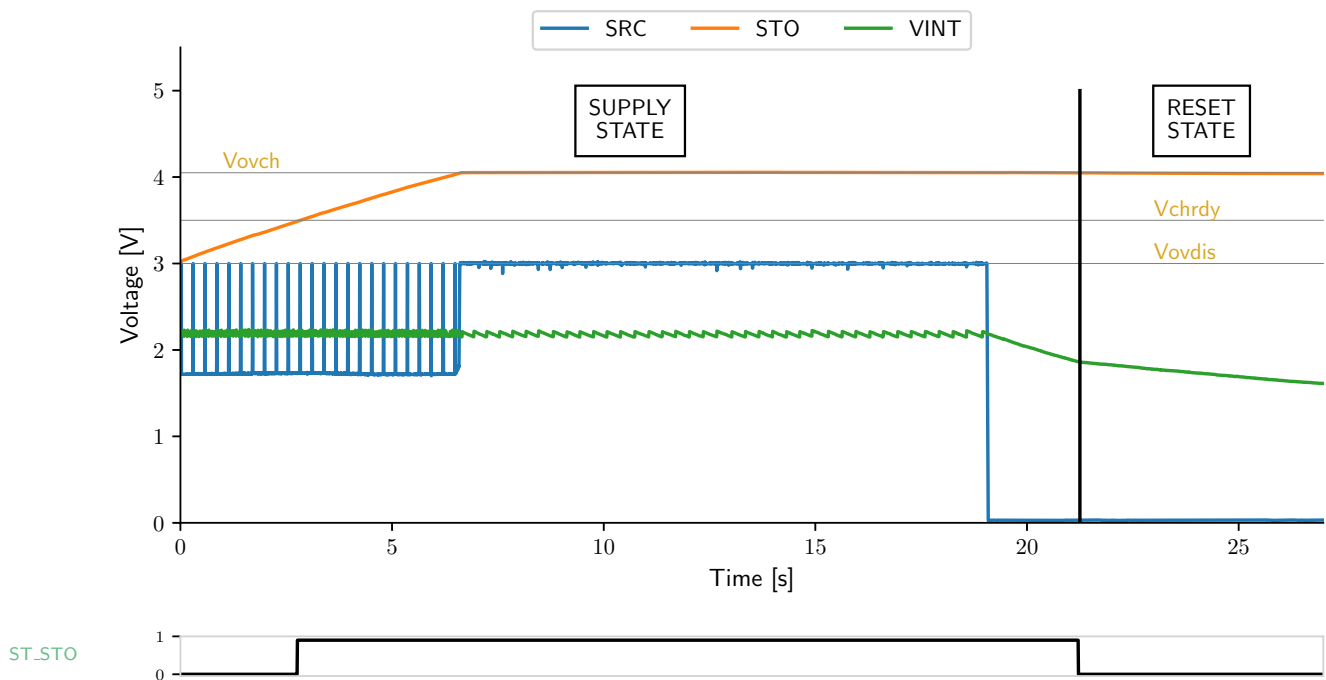


Figure 15: Supply state and Reset state

- The AEM10300 is initially in **SUPPLY STATE** with **V_{VINT}** regulated around 2.2 V, **V_{SRC}** regulated to **V_{MPP}** with **V_{OC}** evaluations every **T_{MPPT,PERIOD}** and **V_{STO}** charging.
- **ST_STO** is asserted once **V_{STO}** is above **V_{CHRDY}**.
- Once **V_{STO}** reaches **V_{OVCH}**, the AEM10300 stops charging the storage element and the **SRC** pin is set to high impedance.
- After 19 s, the power supply connect on **SRC** is switched off. The AEM10300 is unable to maintain **V_{VINT}** to 2.2 V.
- Once **V_{VINT}** dropped below to reset voltage (1.8 V), the AEM10300 goes into **RESET STATE**. **ST_STO** is set LOW.
- The AEM10300 will stay in **RESET STATE** until the minimum required power **P_{SRC,CS}** is available on **SRC** to coldstart again.

9. Schematic

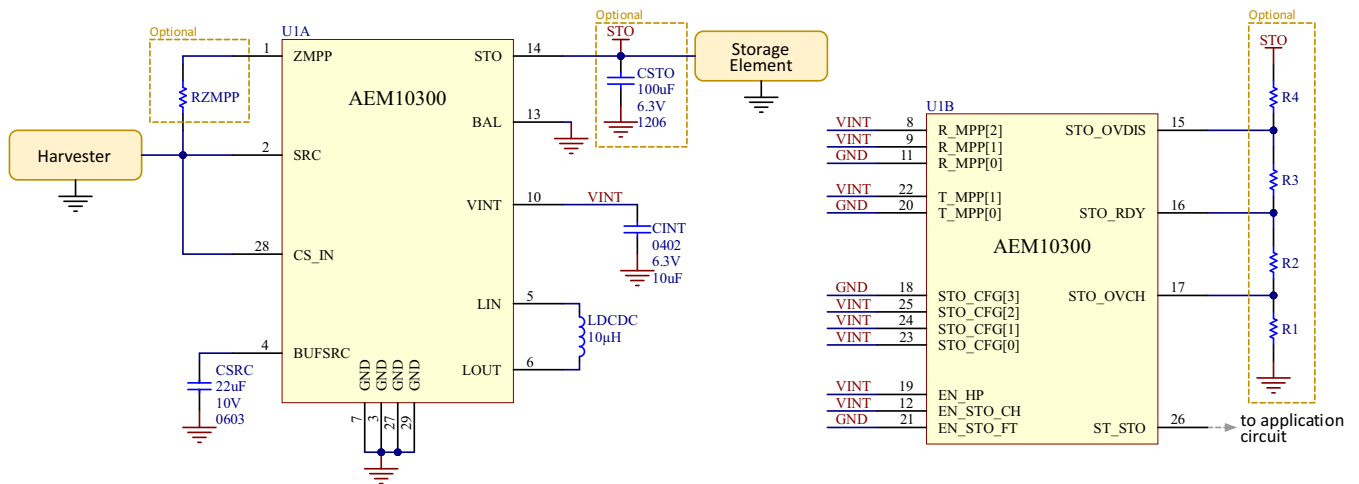


Figure 16: Schematic Example

Designator	Description	Quantity	Manufacturer	Link
U1	AEM10300 - Symbol QFN 28-pin	1	e-peas	order at sales@e-peas.com
L _{DCDC}	Power inductor 10 μ H - 1.76A	1	Murata	DFE252010F-100M
C _{INT}	Ceramic Cap 10 μ F, 6.3V, 20%, X5R 0402	1	Murata	GRM155R60J106ME15
C _{SRC}	Ceramic Cap 22 μ F, 10V, 20%, X5R 0603	1	Murata	GRM188R61A226ME15
C _{STO} (optional)	Ceramic Cap 220 μ F, 6.3V, 20%, X5R 1206	1	Murata	GRM31CR60J227ME11

Table 12: Minimal Bill of Materials

10. Layout

10.1. Guidelines

Good layout practices are mandatory in order to obtain good stability and best efficiency with the AEM10300. It also allows for minimizing electromagnetic interferences generated by the AEM10300 DCDC converter.

The following list, while not exhaustive, shows the main attention points when routing a PCB with the AEM10300:

- The switching nodes (LIN and LOUT) must be kept as short as possible, with minimal track resistance and minimal track capacitance. Low resistance is obtained by keeping track length as short as possible and track width as large as possible between L_{DCDC} and the AEM10300 pins. Minimal capacitance is obtained by keeping distance between LIN/LOUT and other signals. We recommend removing the ground plane, the power plane and the bottom layer ground pour under L_{DCDC} footprint, as well as adding distance between LIN/LOUT and the top ground pour, as shown on Figure 17.
- The DCDC decoupling capacitors (C_{SRC} - C_{STO}) must be placed as close as possible to the AEM10300, with direct connection and minimum track resistance for the corresponding power nodes (BUFSRC and STO).
- The GND return path between the DCDC decoupling capacitors (C_{SRC} - C_{STO}) and the AEM10300 thermal pad, which is the AEM10300 main GND connection, must be as direct and short as possible. This is preferably done on the top layer when possible, otherwise by internal/bottom plane, using low resistance vias to decrease layer-to-layer connection resistance.
- The external DC power connections (SRC and STO) must be connected to the AEM10300 with low resistance tracks.
- Connection between VINT and C_{INT} must be moderately short for AEM10300 stability, even though this pins does not carry large currents. Same for connection between C_{INT} to GND.
- If used, ZMPP must be connected to the AEM10300 with a low resistance track, according to the expected SRC power.
- The BAL pin connection track must be able to handle at least 40 mA.
- The custom mode setting pins STO_OVDIS, STO_RDY and STO_OVCH are high impedance analog inputs typically connected to a resistive divider with high resistor values, making those three nodes prone to pickup noise. Thus, it is recommended to keep those as short as possible and as far as possible to noise sources such as DCDC switching nodes.
- The configuration pins and the status pins have minimal layout restrictions. CS_IN maximum current is below 1 mA, so its layout restrictions are minimal as well.

10.2. Layout Example

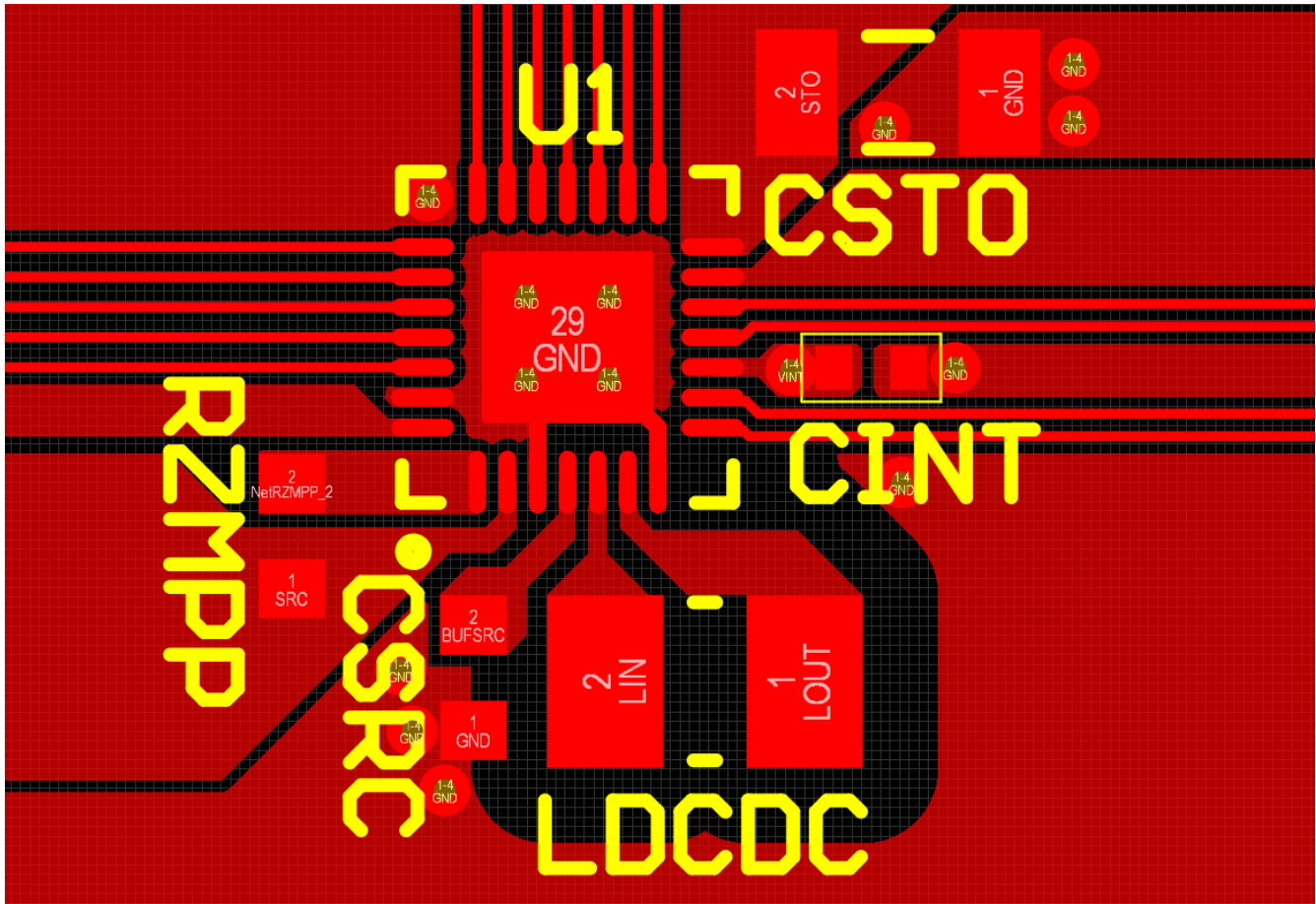


Figure 17: Layout Example for the AEM10300 and its Passive Components

NOTE: schematic, symbol and footprint for e-peas component can be ordered by contacting e-peas support team at support@e-peas.com

11. Package Information

11.1. Moisture Sensitivity Level

Parameter	Moisture sensitivity level (MSL)
QFN 28-pin - 4x4mm	Level 1, according to JEDEC-A113 standard

Table 13: Moisture sensitivity level

11.2. RoHS Compliance

e-peas product complies with RoHS requirement.

e-peas defines “RoHS” to mean that semiconductor end-products are compliant with RoHS regulation for all 10 RoHS substances.

This applies to silicon, die attached adhesive, gold wire bonding, lead frames, mold compound, and lead finish (pure tin).

11.3. REACH Compliance

The component and elements used by e-peas subcontractors to manufacture e-peas PMICs and devices are REACH compliant. For more detailed information, please contact e-peas sales team.

11.4. Tape and Reel Dimensions

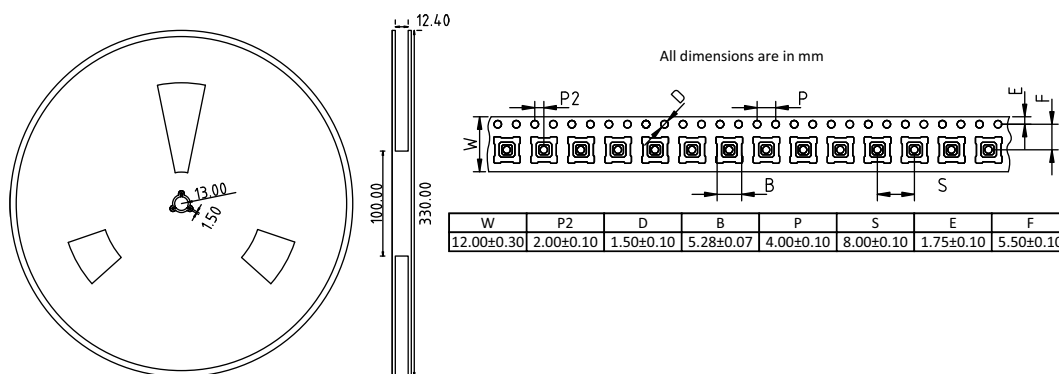


Figure 18: Tape and reel dimensions

11.5. Package Dimensions

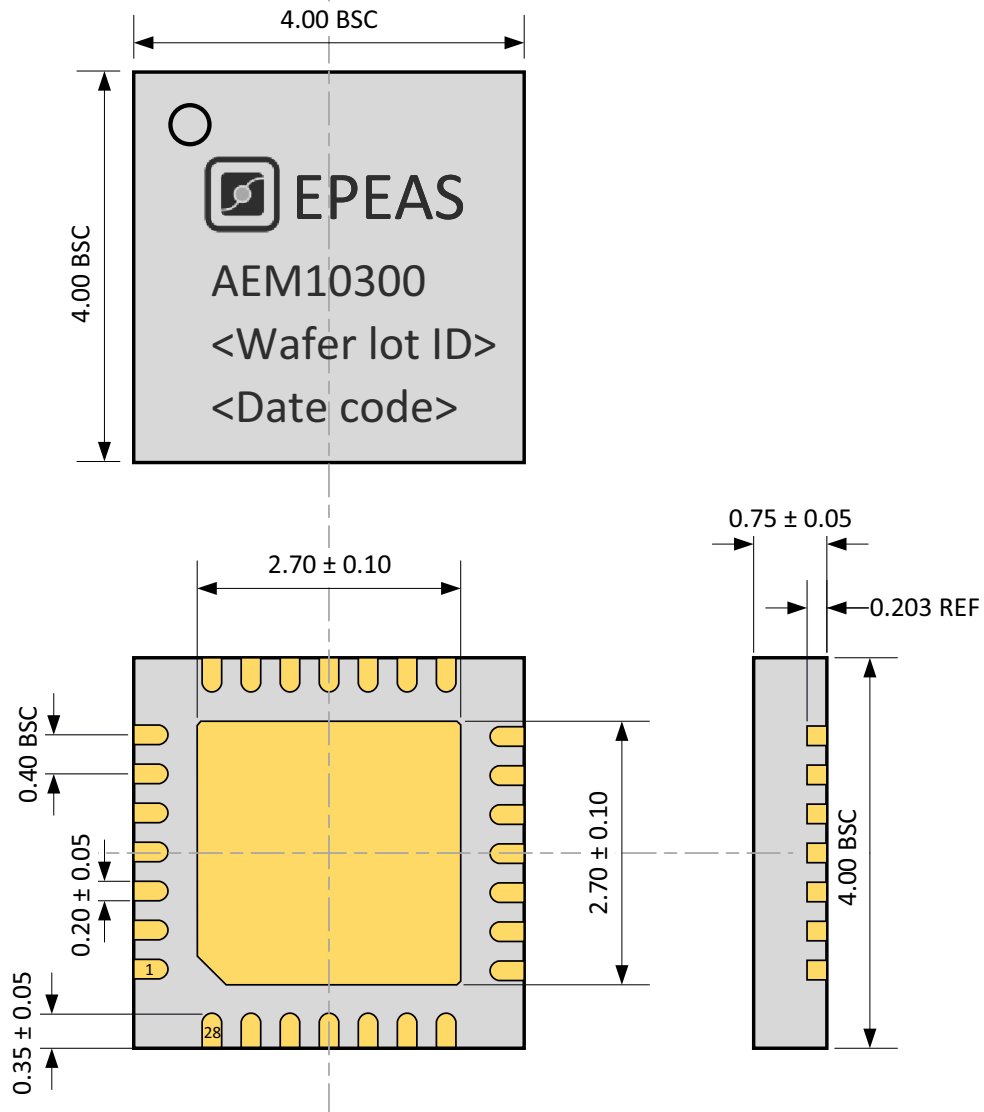


Figure 19: QFN 28-pin 4x4mm Drawing (All Dimensions in mm)

11.6. Board Layout

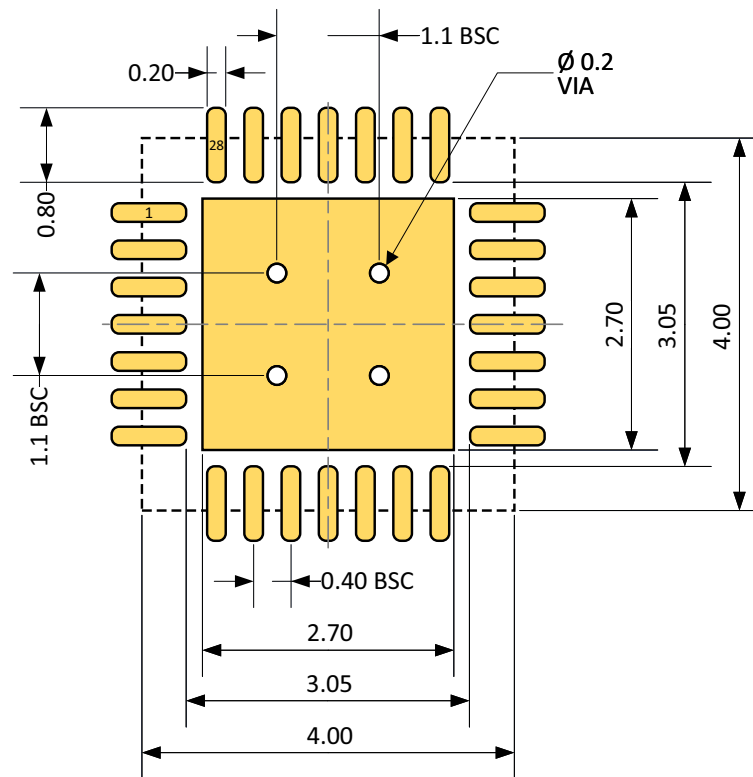


Figure 20: Recommended Board Layout (All Dimensions in mm)

12. Glossary

C_{INT}

Decoupling capacitor on **VINT** pin.

C_{SRC}

Decoupling capacitor on **BUFSRC** pin.

C_{STO}

Decoupling capacitor on **STO** pin.

L_{DCDC}

DCDC converter inductor.

I_Q

Quiescent current pulled from **STO** pin.

$P_{SRC,CS}$

Minimum source power required to coldstart the AEM10300.

MPP

Maximum Power Point.

MPPT

Maximum Power Point Tracking.

R_{MPPT}

For the DCDC converter, ratio between the open circuit voltage V_{OC} and the voltage regulation V_{MPP} determined by the MPPT (see Section 5.4 and Section 6.5).

R_{ZMPP}

Resistor used for ZMPP module, connected between **SRC** and **ZMPP** (see Section 6.6).

R_T

Sum of resistors to set the battery protection threshold voltages in custom mode. $R_T = R1 + R2 + R3 + R4$.

$T_{GPIO,MON}$

GPIO monitoring rate.

$T_{MPPT,WAIT}$

Time during which the AEM10300 stops pulling current from **SRC** to let the source voltage rise to V_{OC} .

$T_{MPPT,MEASURE}$

Duration of the harvester open-circuit voltage (V_{OC}) measure.

$T_{MPPT,PERIOD}$

Period of the MPPT module V_{OC} evaluations (see Section 5.4 and Section 6.5).

$T_{STO,MON}$

STO monitoring rate.

V_{BAL}

Voltage on **BAL** pin.

$V_{BAL,MIN}$

Minimum voltage on the **BAL** pin for the balancing circuit to operate.

V_{OVDIS}

Minimum voltage accepted on the storage element before resetting **ST_STO**.

V_{CHRDY}

Minimum voltage required on the storage element before asserting **ST_STO**.

V_{OVCH}

Maximum voltage accepted on the storage element before disabling its charging.

V_{VINT}

Voltage on **VINT** pin.

V_{SRC}

Voltage on **SRC** pin.

$V_{SRC,CS}$

Minimum source input voltage required to coldstart the AEM10300.

V_{STO}

Voltage on **STO** pin.

V_{MPP}

Target regulation voltage of the DCDC **SRC** input, as determined by the MPPT module.

V_{OC}

Open-circuit voltage of the harvester connected to **SRC**.

13. Revision History

Revision	Date	Description
1.0	June, 2021	First version of the document
1.1	August, 2021	Minor modifications
1.2	March, 2023	- Various aesthetic improvements. - Explanations about BAL circuit. - Added EN_STO_FT in Configuration Pins table - Section with precisions about the use of CS_IN. - Fixed state machine graph. - Fixed footprint dimensions - New "behavior" oscilloscope graphs with improved description. - Moved various states description sections as sub-sections of a global section. - Supply State description: explanation about SRC being set to high impedance when all nodes are fully charged. - Updated "Typical use" of storage element vs STO_CFG[3:0] configuration. - Replaced "asserted/de-asserted" by "HIGH/LOW". - Changed CSRC from 15μF/0402 to 22μF/0603. - Updated "Recommended Operation Conditions" with minimum capacitor values including derating and tolerances.
1.3	November, 2023	- Updated efficiency graphs. - Created section for pinout. - Updated schematics with new symbol. - Digital levels High/Low: replaced 0/1 notation by L/H. - Fixed example circuits errors. - Fixed typos and aesthetic issues. - Added layout guidelines with clearer layout examples.
1.4	February, 2024	Fixed wrong HIGH level of ST_STO in "Power and Status Pins" table.
1.5	July, 2026	- Reworked the whole datasheet. - Added REACH and RoHS compliances. - Added GPIOs and STO monitoring rates. - Renamed the MPPT timings to $T_{MPPT, WAIT}$, $T_{MPPT, MEASURE}$, and $T_{MPPT, PERIOD}$. - Fixed confusion between R_{ZMPP} and R_{MPPT} in "Typical Electrical Characteristics" table. - Added $V_{BAL, MIN}$ and improved balancing circuit explanations. - Corrected SUPPLY STATE explanations when V_{STO} falls below V_{OVDIS}. - Corrected "Typical Application Circuits" figures to match the descriptions. - Added cautionary statement along with storage element threshold voltages. - Added tape and reel dimensions. - Updated package and recommended board layout figures. - Added glossary.

Table 14: Revision History